LINUX BOOT PROCESS

Power On

BIOS/UEFI Initialization

- When you power on your computer, the BIOS or UEFI firmware is loaded from non-volatile RAM (NVRAM).
- The BIOS/UEFI is responsible for initializing hardware components, performing a Power-On Self-Test (POST), and starting the boot process.

Probe for Hardware

- The BIOS/UEFI probes or detects the hardware components connected to the system, including the CPU, memory, storage devices, and peripherals.

Select Boot Device (HDD, USB, PXE,...)

- After hardware detection, the BIOS/UEFI allows you to select the boot device from which the operating system will be loaded.
- You can choose from options like booting from a local disk, a network server, or other storage media.

Identify EFI System Partition

- If the system is using UEFI firmware, it identifies the EFI System Partition (ESP) on the boot device.
- The ESP contains boot loaders and other essential boot-related files.

Load Boot Loader (e.g., GRUB2)

- The BIOS/UEFI or UEFI firmware loads the chosen boot loader. In many Linux systems, GRUB (Grand Unified Bootloader) is commonly used as the boot loader.
- GRUB provides a menu to select the operating system to boot or automatically loads the default Linux kernel.

Determine Which Kernel to Boot

- GRUB or the chosen boot loader determines which Linux kernel to load.
- This decision is typically based on the kernel's version and configuration specified in the boot loader's configuration files.

Load the Selected Kernel

- The boot loader loads the selected Linux kernel into memory.
- The kernel is the core of the operating system and is responsible for hardware initialization and managing system resources.

Instantiate Kernel Data Structures

- After loading, the kernel initializes its data structures, sets up memory management, and prepares for the transition to user space.

Start Init/systemd as PID1

- The kernel starts the init system or systemd as the first user-space process (PID 1).
- In modern Linux distributions, systemd has become the default init system, responsible for managing system services and processes.

Execute Startup Scripts

- The init system or systemd executes startup scripts and initializes system services and daemons.
- These scripts and services include those responsible for setting up networking, mounting filesystems, and other system-level tasks.

Running System

Once all initialization and startup tasks are completed, the Linux system is fully booted and ready to use.

Users can log in, and the system is in a running state, allowing users to run applications and perform tasks.